Complex percolation on the doubly-principal cylinder sigma-algebra

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Abstract

For complex transitive percolation, a finite countably additive complex measure matching the prescribed naturally labelled doubly-principal cylinder amplitudes exists exactly when the percolation parameter belongs to $[0, 1]$. The obstruction already appears on the sigma-algebra generated by these cylinders, without using all finite cylinders. Two universal caps reduce the absolute level sum to a weighted sum over arbitrary finite orders. Normalizing the modulus weights gives a real random graph order and its standard endpoint-connectivity product, yielding an exponentially divergent lower bound outside the real segment. This completes a calculation suggested in the direct predecessor, rather than introducing a general extension criterion. A separate covariant result requires an all-principal common-factor rule and normalization on infinitely many posts. A Dirac counterexample shows why that rule cannot be weakened to doubly-principal pasts alone.

1 The labelled classification

Let $[n] = \{0, \dots, n-1\}$. A naturally labelled finite order C on $[n]$ has $i \prec_C j$ only if $i < j$. Write $L(C)$ for its number of covers, $R(C)$ for its number of strict comparable pairs, and $h(C) = \binom{n}{2} - R(C)$ for its number of incomparable pairs. For $p \in \mathbb{C}$ and $q = 1 - p$, prescribe

$$A_p(C) = p^{L(C)} q^{h(C)}. \tag{1}$$

Powers with exponent zero equal one, also at $p = 0$ or $q = 0$.

The space $\widetilde{\Omega}$ consists of orders on $\mathbb{N}$ with the same natural-label property. Its usual sigma-algebra $\widetilde{\mathcal{F}}$ is generated by finite-prefix cylinders $\mathrm{cyl}(C) = \{\omega : \omega|_{[n]} = C\}$. An order is *principal* if it has a unique maximal element. For $n \geq 2$, it is *doubly principal* if both C and $C|_{[n-1]}$ are principal. Set

$$\widetilde{\mathcal{D}} = \sigma\{\mathrm{cyl}(C) : C \text{ is doubly principal}\} \subseteq \widetilde{\mathcal{F}}. \tag{2}$$

We write $\widetilde{\mathcal{D}}$ to avoid using a numerical parameter as the subscript for an algebra that is independent of p. This is the labelled algebra denoted $\widetilde{\mathcal{R}}_p$ in Dowker–Zalel [1, Sec. V].

Theorem 1 (Labelled doubly-principal classification)**.** *There exists a countably additive complex-valued measure $\mu : \widetilde{\mathcal{D}} \to \mathbb{C}$ such that $\mu(\mathrm{cyl}(C)) = A_p(C)$ for every doubly-principal C if and only if $p \in [0, 1]$. The same existence classification holds if $\mu(\widetilde{\Omega}) = 1$ is additionally required.*

A complex-valued measure here takes only finite complex values and is countably additive on its sigma-algebra. Such a measure has finite total variation. In particular, for any finite disjoint measurable family,

$$\sum_i |\mu(E_i)| \leq |\mu|(\widetilde{\Omega}) < \infty. \tag{3}$$

This is a standard property of scalar complex measures, not an extra positivity assumption; compare the bounded-variation framework in [4, Sec. 2.3 and App. B].

1.1 Two elementary identities

For an arbitrary naturally labelled order G of size k, let $M(G)$ be its number of maximal elements and put $w(G) = L(G) + M(G)$. For the empty order all these counts are zero. The ordinal sum $G \oplus \{u\} \oplus \{v\}$ adds two universal maxima, in that order.

Lemma 2 (Two caps)**.** *The map $G \mapsto C_G = G \oplus \{u\} \oplus \{v\}$ is a bijection from naturally labelled k-orders to doubly-principal $(k+2)$-orders, and*

$$L(C_G) = L(G) + M(G) + 1, \qquad h(C_G) = h(G), \qquad w(G) \geq k. \tag{4}$$

Proof. In a finite order a unique maximal element is above every other element. Natural labelling places it last. Applying this observation twice gives the unique two-cap decomposition and its converse. The first cap covers exactly the maximal elements of G; the second covers just the first cap. Neither adds incomparable pairs. Every nonmaximal vertex of G is the lower endpoint of a cover. Distinct lower endpoints give distinct covers, so $L(G) \geq k - M(G)$. □

Lemma 3 (Endpoint product)**.** *If $0 < r, s < 1$ and $r + s = 1$, then*

$$\sum_{G:\,|G|=k} r^{w(G)} s^{h(G)} = F_k(s) := \prod_{j=1}^{k} (1 - s^j) \geq c_s, \qquad c_s := \prod_{j=1}^{\infty} (1 - s^j) > 0. \tag{5}$$

The sum is over distinct naturally labelled orders, without dividing by automorphisms.

Proof. Independently select each forward edge on $k+1$ ordered vertices with probability r and take transitive closure. For a prescribed closure C, all its cover edges must occur and edges joining its incomparable pairs must be absent. Every remaining comparable edge is redundant and its two choices sum to $r + s = 1$. Thus

$$\Pr(C) = r^{L(C)} s^{h(C)}. \tag{6}$$

The event that the last vertex is above all earlier vertices consists of exactly the orders $G \oplus \{u\}$, and therefore has probability equal to the sum in (5). Reverse the vertex order and all arrows. The edge law is unchanged and the event becomes reachability of every later vertex from the first. Once the first vertex reaches the j vertices already present, it reaches the next vertex precisely when at least one of its j incoming edges occurs. These edges have not previously been exposed, giving the successive factors $1 - s^j$.

This is the standard one-sided product used in random-graph post probabilities [5, Sec. 2, Eq. (2)]; the argument is included to fix the indexing. Finally $\sum_j s^j < \infty$. For sufficiently large j, $\log(1 - s^j) \geq -2s^j$, so the tail product has a positive limit. The finite initial product is positive too. □

1.2 Variation lower bound and proof of Theorem 1

For $p \notin [0, 1]$, put

$$a = |p|, \quad b = |1 - p|, \quad t = a + b, \quad r = a/t, \quad s = b/t. \tag{7}$$

Both a, b are positive. Equality in $|p| + |1-p| \geq 1$ holds exactly for $p \in [0,1]$, so $t > 1$. Let

$$V_n(p) = \sum_{\substack{C \text{ doubly principal} \\ |C|=n}} |A_p(C)|. \tag{8}$$

By Lemmas 2 and 3,

$$\begin{aligned} V_{k+2}(p) &= a \sum_{|G|=k} a^{w(G)} b^{h(G)} \\ &= a \sum_{|G|=k} t^{w(G)+h(G)} r^{w(G)} s^{h(G)} \\ &\geq a t^k F_k(s) \geq a t^k c_s. \end{aligned} \tag{9}$$

Here $w(G) + h(G) \geq k$ and $t > 1$; no equality is asserted at the first inequality. In particular,

$$\boxed{V_n(p) \geq |p|\big(|p| + |1-p|\big)^{n-2} c_s \longrightarrow \infty.} \tag{10}$$

Proof of Theorem 1. For each fixed n, the cylinders counted in V_n are disjoint and belong to $\widetilde{\mathcal{D}}$. Any matching complex measure would make $V_n \leq |\mu|(\widetilde{\Omega})$ for every n by (3), contradicting (10). This rules out even unnormalized measures outside $[0,1]$.

For real $0 < p < 1$, the countable independent-forward-edge probability space pushes forward under transitive closure to a probability on $(\widetilde{\Omega}, \widetilde{\mathcal{F}})$. The closure calculation (6) gives precisely $A_p(C)$ for every cylinder. Restrict this probability to $\widetilde{\mathcal{D}}$. At $p = 0$ use the Dirac probability on the infinite antichain; at $p = 1$ use the Dirac probability on the infinite chain. They have the prescribed endpoint values, with the zero-power convention stated above, and remain normalized upon restriction. □

Corollary 4. *The same nonexistence result holds on any sigma-algebra containing $\widetilde{\mathcal{D}}$ if the doubly-principal values are retained, in particular on the principal cylinder sigma-algebra and on $\widetilde{\mathcal{F}}$.*

Proof. A matching measure on a larger domain would restrict to the measure excluded by Theorem 1. □

The converse inference is invalid: nonexistence on a larger domain alone would not prove Theorem 1.

2 A separate conditional statement for covariant events

Let $\mathcal{R} \subseteq \widetilde{\mathcal{F}}$ be the sigma-algebra of measurable events invariant under order-isomorphism of the infinite histories. A *post* is an element comparable with every other element. Let $\mathcal{P}_\infty$ be the event of infinitely many posts, and define

$$\mathcal{R}_{\text{post}} = \{E \in \mathcal{R} : \mathcal{P}_\infty^c \subseteq E \text{ or } \mathcal{P}_\infty^c \subseteq E^c\}. \tag{11}$$

This is the covariant posts algebra of [1, Sec. V, Eq. (29)]. It is not the labelled algebra $\widetilde{\mathcal{D}}$.

For each finite principal unlabelled shape H, let D_H be the event that the history belongs to $\mathcal{P}_\infty$ and has a post whose inclusive past is isomorphic to H. Let $N(H)$ be the number of

distinct naturally labelled representatives of H; labellings related by an automorphism are already identified in this count. The amplitude $A_p(H)$ is well defined by its order-invariant exponents. Impose the following additional rule for *all* principal shapes:

$$\nu(D_H) = \kappa N(H) A_p(H), \qquad \kappa \in \mathbb{C} \text{ independent of } H \text{ and } |H|. \tag{T}$$

This is a model assumption about values, not a consequence of covariance.

Theorem 5 (Conditional common-factor classification)**.** *A countably additive complex-valued measure ν on $\mathcal{R}_{\mathrm{post}}$ with $\nu(\widetilde{\Omega}) = \nu(\mathcal{P}_\infty) = 1$ and a finite κ satisfying* (T) *exists if and only if $0 < p \le 1$.*

Proof. First, D_H is measurable: a post with $n-1$ ancestors must have label $n-1$ in every natural labelling. Its finite past is a finite cylinder condition and persistence as a post is a countable intersection of finite conditions. Intersect with the measurable event $\mathcal{P}_\infty$. The result is isomorphism-invariant and excludes $\mathcal{P}_\infty^c$, so belongs to $\mathcal{R}_{\mathrm{post}}$.

At fixed n, different shapes H give disjoint events. Across ranks the family is laminar: if $|H| < |J|$, then $D_H \cap D_J$ is empty or D_J. Indeed, the finite past J of the later post determines whether an earlier post of the indicated rank and shape exists inside J. Everything beyond the later post lies above the earlier one as well. The countable family $\{D_H\}$ covers $\mathcal{P}_\infty$.

If $\kappa = 0$, every family member has value zero. Its finite intersections are empty or family members; inclusion–exclusion makes each finite union zero. Continuity from below for finite complex measures then gives $\nu(\mathcal{P}_\infty) = 0$, a contradiction. Thus $\kappa \neq 0$. This argument uses laminarity; a union of arbitrary complex-measure zero sets need not have value zero.

For $p \notin [0,1]$, regroup the labelled sum in (8) by unlabelled doubly-principal shapes. Natural-label multiplicities are exactly $N(H)$, so the disjoint rank-n events satisfy

$$\sum_{\substack{[H] \text{ doubly principal} \\ |H|=n}} |\nu(D_H)| = |\kappa| \sum_{[H]} N(H) |A_p(H)| = |\kappa| V_n(p). \tag{12}$$

This diverges by (10), contradicting finite total variation. At $p = 0$, every principal H with $|H| \ge 2$ has a cover, so its amplitude vanishes. Those sizes alone still cover $\mathcal{P}_\infty$; the laminar argument again contradicts normalization.

For $0 < p < 1$, classical transitive percolation has infinitely many posts almost surely, a prior result recorded in [5, Sec. 2] and attributed there to Alon et al. Its restriction to $\mathcal{R}_{\mathrm{post}}$ has (T) with

$$\kappa = c(1-p), \qquad c(q) = \prod_{j \ge 1} (1 - q^j) > 0. \tag{13}$$

The finite past has probability $N(H)A_p(H)$. Conditional on that past, the top remains below every later birth with probability $c(q)$: after it has reached j vertices, the next birth connects to at least one of them with probability $1 - q^j$. This future factor is independent of the shape of H. The argument is the classical post product [5, Theorem 2.1]. At $p = 1$ the infinite chain gives the required measure with $\kappa = 1$. □

Proposition 6 (Why all principal shapes are required)**.** *If* (T) *is required only for doubly-principal H, a normalized measure satisfying it with $\kappa = 0$ exists for every numerical p.*

Proof. Take the infinite ordinal sum

$$(A_2 \oplus \{z_1\}) \oplus (A_2 \oplus \{z_2\}) \oplus \cdots, \tag{14}$$

where A_2 is a two-element antichain. Each z_j is a post. The other elements are incomparable with their partner, hence are not posts. Deleting z_j from its inclusive past leaves two maximal elements, so no inclusive post past is doubly principal. The Dirac measure at this history, restricted to $\mathcal{R}_{\mathrm{post}}$, has $\nu(\widetilde{\Omega}) = \nu(\mathcal{P}_\infty) = 1$ and assigns zero to every doubly-principal D_H. It satisfies exactly the weakened rule. □

3 Prior work and the size of the contribution

Three predecessors locate the mathematical delta. *Dowker–Zalel* [1, Sec. V, Eqs. (29)–(31)] defines the target algebra and leaves the doubly-reduced extension test inconclusive. Its closing paragraph already suggests using post-probability techniques for a cumulative doubly-principal amplitude sum. The individual level sum V_n here is not literally that cumulative quantity. The contribution is the explicit bound (10) and the resulting classification, executing the suggested direction. *Zalel's thesis* [2, Ch. 6, Sec. 6.2.3, pp. 89–90] records the earlier super-reduced formulation and defers the calculation. The thesis read depth is inherited from the prior source-record lane; this drafting task did not reopen it, and the load-bearing priority statement is the published Dowker–Zalel passage above. *Dowker–Johnston–Surya* [3, Sec. 4] already establishes full-labelled complex-percolation nonextension; Surya–Zalel [4, Theorem 2.1] states the scalar bounded-variation extension criterion. Failure on those larger domains does not by itself settle this smaller prescribed-value problem.

The endpoint product is established probability theory, rederived in Lemma 3. No new extension criterion or probability method is claimed. The conditional statement is additional only under its stated value rule; arbitrary covariant measures are not classified. Targeted project priority searches and independent source checks found no matching proof at their recorded depths. They are not an exhaustive literature search or a certificate of global priority. A prior proof of the same prescribed-value classification would remove the publication delta without invalidating the argument.

4 Scope and falsifiers

Scalar amplitudes versus quantum measures. Countable additivity concerns the complex amplitude measure μ. The set function $Q(E) = |\mu(E)|^2$ is generally not additive. The theorem does not apply the probability sum rule to Q, nor classify arbitrary decoherence functionals. Generalized event-convergence prescriptions or finitely additive assignments are outside its hypotheses.

An infinite-dimensional norm-only escape. At $p = (1+i)/2$, let $P_{1/2}$ be classical percolation and define $m(E) = 1_E \in L^2(P_{1/2})$. This standard indicator-valued measure is countably additive in norm, since the squared norm of a disjoint tail is its probability. On every finite cylinder,

$$\|m(\mathrm{cyl}(C))\|_2^2 = P_{1/2}(\mathrm{cyl}(C)) = |A_p(C)|^2.$$

Thus matching cylinder norms in a new Hilbert space is possible even where the prescribed complex measure does not exist. Its vector total variation is infinite by (10). It preserves neither the scalar phases nor the original off-diagonal decoherence data. It is a scope counterexample, not an extension of the fixed complex premeasure or a proposed dynamics.

Concrete ways the argument could fail. A counterexample to the two-cap bijection or the cover count in (4), a failure of the endpoint identity (5), or a matching finite complex measure with

unbounded disjoint sums would defeat the main proof. For Theorem 5, an overlap contradicting laminarity or a missing natural-label multiplicity would defeat its additional step. Finite checks below test these encodings; the proofs supply their all-level justification. Rank-dependent factors and generalized regularizations are not asserted as theorems in this note.

Physical and philosophical limits. The event domain, scalar codomain, countable additivity and matching rule are mathematical model choices. No observer-access model, empirical prediction or interpretation of growth as objective passage is supplied. A conclusion about temporal ontology would require independent physical and philosophical bridge premises. Neither this theorem nor a finite formal certificate supplies them.

A Reproducibility and finite formal scope

The accompanying repository contains the exact-arithmetic runner `research/paper-novel/reproduce.py`. From the repository root run

```
python3 research/paper-novel/reproduce.py
```

It records the selected source revision and hashes, runs the accepted independent finite scripts without changing their outputs, and exits nonzero on failure. It verifies the two-cap values at $p = (1+i)/2$,

$$V_2 = \frac{1}{\sqrt{2}}, \qquad V_5 = \frac{3}{4} + \frac{3\sqrt{2}}{16}, \qquad V_6 = \frac{31}{32} + \frac{55\sqrt{2}}{64},$$

along with 407 exact small-cylinder identities through five labels and the rank-one counterassignment that prevents an incorrect local-renewal extension. The GLM review suite independently checks 26 finite identities, including natural-label multiplicities and an older ordinal-block certificate.

No single finite sum exceeding one proves nonextension: an unknown finite complex measure need not have total variation one. The contradiction is the *unbounded sequence* (10). Enumerated cases do not establish its infinite-product or all-level steps.

An existing light Lean certificate in `formal/lean/Certificate.lean` checks a finite numerical kernel. Its recorded axiom audit does not formalize sigma-algebras, countable additivity, the positive infinite product, or either classification theorem. This manuscript does not present those theorems as machine-verified. AI-assisted derivation and review were used; the mathematical evidence is the written proof, source correspondence and reproducible checks, not agreement among agents.

The accompanying `CLAIM_LEDGER.md` separates original calculations, prior results, conditional assumptions and bounded checks, with exact source locators and read-depth limitations. `README.md` gives build commands and the manuscript's review status.

References

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